

Homework #1

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QUESTION 1

1.1

One activity that I perform with a computer interface in multiple different contexts is scrolling on TikTok while walking on the street, at the dinner table eating, and while sitting on the couch.

1.2

While walking on the street, the constraints are that my cognitive abilities are limited due to needing to pay attention to my external environment, so my cognitive resources are divided therefore significantly reducing the attention I must be able to scroll. For example, when crossing the street at a crosswalk, cars are often turning right or left so I must be extra aware that they don't accidentally run me over in the case they don't see me which physically pulls my attention away from scrolling on TikTok. I also must keep in mind that I might bump into something or someone else in my physical environment that really limits my cognitive ability to focus on scrolling on TikTok because I must be aware of physical objects that might harm me.

While sitting on the couch, they are very little. The reason is because I am in a physically fixed position where my directed attention, both physically and mentally, is fixated on scrolling on TikTok. The only constraint I imagine myself or another user facing is the fact that while you're on your phone, you have a number of different distractions pop up like notifications that might limit your attention, which more often than not, does happen to me, for example, when I see a message from my mom saying that dinner is ready.

While sitting down at the dinner table eating, the constraints are something in the middle between both the above contexts of while walking and sitting on the couch. While I'm actively eating, it's a lot easier to scroll than when you're walking, but not as easy as it is if I'm solely focused on scrolling when sitting on the couch where that's my primary focus. Scrolling while eating physically involves

me having my phone in one hand, or propped up but still having to swipe, and then one hand eating. My cognitive focus is physically limited from scrolling on TikTok in order to pick up the food I'm eating as well as less precise due to having to scroll with one hand and eat with the other hand. However, while I physically am not watching, I'm audibly still absorbing what's going on in the video helping me infer what is going on visually. This allows me to create a mental model of what's happening in between that moment where I look down and back up. Often, this is almost automatic, and I've gotten pretty good at multi-tasking between eating without even looking down and scrolling and if I do look down, I time it so there's not much going on in the video I'm watching that would recruit much cognitive focus or I pause from scrolling in that moment.

1.3

While walking on the street, the interface can be designed to overcome constraints by limiting controls of the app to what are necessary such as auto pausing when a person stops walking because their attention is briefly directed to their physical environment and not on their phone anymore. Cognitive and physically demanding features like a *'comment'* can be temporarily disabled. Haptic feedback at the end of a video to signal it done in case a user is looking away from the video but still ingesting the content audibly is another alternative. These changes can assist in allowing the user to use the interface while in the context of walking which might reduce their physical senses and cognitive focus on the activity of scrolling.

While sitting on the couch, the interface can be designed to overcome constraints to have richer interactions because the interface knows the user has most of their cognitive power on the activity of scrolling at hand. For this, actions like tapping to expand features on comments/threads and ensuring all controls are fully enabled. The design could also limit notifications or unnecessary interruptions like notifications of messages from other people, being tagged on a post, comment, or thread so as not to distract the user from the activity of scrolling.

While eating at the dinner table, the interface can be designed to by optimizing the actions users do to achieve their goals that would need usually need two hands into one. Actions like typing a comment, searching for a video or topic, navigating the menus, could be designed for single swipe-based controls or voice input where users can speak commands like *'rotate my screen'* or *'zoom in on this*

comment'. Similar to the walking context, clearer cues for when a video is done can be implemented like haptic feedback in the case that a user is looking down to grab some food or away briefly for any reason.

QUESTION 2

2.1

At the top left of the UI, a *'New Thread'* button is presented for posting a thread in which a user infers would be where they would go to ask a question. The system opens a UI that takes up a majority of the screen to fill out their post specifications. It prompts the user to fill out their question content specifications

Once a user is ready to post, they are presented with a *'post'* button at the top right of the post's UI and another at the bottom of the screen. After doing so, it automatically replaces the UI with the post the user generated, pushes their thread to the top of the discussion board, and adds them as a watcher to the post which they can use to filter posts in the discussion panel.

From there, the user waits for a response to be generated, either from another student or someone from the staff. Once a response is generated, the student can either choose to reply or not depending on what they are ultimately feeling regarding their intentions. They may feel that they need to reply to someone's response by pressing *'comment'* under someone's response and use that to message them back and forth until they are satisfied with the answer. They might also react to someone else's answer by pressing the *'heart'* button to signify they are liking their response. They can also check the *'checkmark'* on someone's answer to signify that they approve of this answer which signifies the author is satisfied with the answer.

2.2

Intention:

A student's intention for using the Ed discussion board is to get an answer to a question they might have. Ed carries the student across that stage by offering them the *'New Thread'* feature that allows them to perform a set of actions that will allow them to get feedback from other users on the platform. From there, they will be able to deduce whether other users' responses answer their question

or not. Assuming the student doesn't know what to do, the system helps them figure it out by having them infer that '*New Thread*' is what will allow the user to ask a question.

Action:

To bridge the gap of the gulf of execution and take them across this stage, Ed allows the user to perform certain tasks to get to their end goal. This includes clicking the '*New Thread*' button which opens a UI to fill out their post specifications. As the user types in a title, Ed displays similar existing threads users can reference to have more context in their post.

Execution:

Ed provides two ways to post at the top right and at the bottom right of the screen in which immediate feedback is displayed in the list of threads as well as opening the post to the user. Once someone has answered their thread, it signals the user by displaying the answer as a comment. From here, a user can interact with someone else by replying until their desired intention, answering their question or discussing the topic of the post, is met.

2.3

Interface output:

In terms of successfully communicating to the student the outcome of their actions, Ed doesn't do a great job. If a user is not on the post and doesn't see someone else's response, the only visuals that they see is a small '(1 new)' in red text next to their post in the discussion thread of posts. As well as if, and only if, they open the notification side panel, they'll see that someone has responded to them at the top, but it won't actually ever notify me unless I click it.

Interpretation:

Once one of these outputs is noticed, the student understands that their question has some activity on it and can view the thread to see if it answers their question or not.

Evaluation:

Ed is successful at carrying the students across this stage because while they are in the post, the student can visually see the information pertaining to their question and can analyze if it does. If it doesn't, they can continue to probe by interacting with other students until they are met with an answer they are satisfied with. This is indicated by a green 'checkmark' the author of the post triggers to mark that the question has been answered and resolved.

QUESTION 3

3.1

An activity from my regular life that struggles with a large gulf of evaluation is baking using the interface of a conventional oven. While the steps to use the oven are known and intuitive, the interface fails to bridge the gulf of evaluation because it doesn't communicate effectively if my action has accomplished my goal.

After selecting the '*bake*' mode and setting my desired temperature using the arrow buttons, the oven begins preheating but doesn't give me any feedback whatsoever that it's gotten to my desired temperature. Visually, it does show the current temperature on the LED screen, but when it reaches the temperature I chose, it doesn't do anything, so I don't know if it's ready unless I inspect the temperature it currently is at. Due to prior experience, I wait and watch while the oven slowly heats up to my desired temperature, or I come back in five minutes and it's usually just reaching my desired temperature. This wide gulf makes knowing if the oven is ready for me to cook my food that I want to eat ambiguous and fails to provide feedback knowing when it is ready.

3.2

A similar, yet different activity from my regular life that does a much better job at bridging the gulf of evaluation is an electric kettle. The intention behind using the electric water kettle is to warm up some water which I use for instant coffee or brewing some tea. After following the steps to boil the water, I know that the water is boiling when it immediately starts audibly rumbling, the whole kettle as well as the lever I used to put it in the 'boil' setting turns a bright blue color. Since it's clear, I can physically see the water boiling, and I can physically see the steam releasing from the spout of the kettle. I know it's fully done boiling when

the water stops boiling, steam stops releasing, and most importantly, the LED has turned off, and the lever makes a loud 'click' sound.

The gulf here is much narrower compared to that of the conventional oven because I get immediate feedback as soon as I turn on the kettle. There are clear signs of feedback that I get from the kettle that narrows the bridge for the gulf of evaluation compared to the conventional oven where I would have to be inspecting the temperature to make sure it matches what I inputted five minutes prior. These visual and auditory cues that I get from the electric kettle makes the state of the interface a much narrower bridge of evaluation compared to a conventional oven where the feedback is delayed and not as easy to interpret without active monitoring.

3.3

The lessons that could be borrowed from the second activity's interface, the water kettle, to resolve the wide gulf of evaluation in the first activity, the conventional oven, are to use auditory and visual signifiers to signal immediate feedback that the action being performed has been received by the oven and that it's in progress and that the desired action has been accomplished. For example, a unique audible chime or even a voice saying '*temperature rising*' would have sufficed in giving me feedback that the oven is working on increasing the temperature to the one I input. As well as an audible chime or voice saying '*Temperature at X has been set*' would have made the gulf a lot narrower where I got feedback that the temperature of the oven has been set. Visually borrowing from the electric kettle example, some sort of text saying '*in progress*' or a loading screen on the LED would be good feedback in knowing that the temperature increase is in progress as well as some text saying '*temperature increased*' as well as a checkmark on the LED screen or a unique audible chime as feedback to signal that the temperature has risen to my desired number .

QUESTION 4

4.1

I believe that it is ethical for ChatGPT to be used in this way because the benefits of having it as a tool outweigh the risks of not having it. I think the main benefit is the easy access and ability for the user to have a conversation with ChatGPT

that makes them feel safe, secure, validated and supported in regards to their feelings and emotions. One major risk here is that it might make users overly dependent as a source of truth instead of an opinion. This issue arises in the this article, [*“ChatGPT wrote “Goodnight Moon” suicide lullaby for man who later killed himself”*](#) where a man committed suicide using ChatGPT in where this model was more “*confidant*” based and the “*chatbot suddenly pretended to know and love Gordon, understanding him better than anyone else in his life*” in which the model was “*programmed with excessive sycophancy, anthropomorphic features, and memory that stored and referenced user information across conversations in order to create deeper intimacy*” (Belanger, 2026). This is the exact kind of dependency risk ChatGPT needs to address before anyone tries to use it not just as a therapist, but anything regarding human emotion or sentiment. As well as that too much personalization could lead to a deeper emotional bond that could make people feel seen and validated and rely on it too much, such as the case with Gordon.

Constraints and disclaimers that should be in place are explicitly stating that ChatGPT is not diagnosing a user’s situation or feelings, prescribing any sort of treatment, and that it’s not a professional in this matter. It should provide a disclaimer that addresses emotional risk and dependency to make users aware. \ Constraints could be to restrict the conversation or direct the user to a professional to seek help if emotional distress or language is detected.

4.2

An experiment to test the effectiveness of this tool would be to test if therapy-like support would improve a user’s emotional stability, support, and mood when used alongside traditional therapy. The design would consist of participants that currently see their existing therapist consistently and would be measured with self-report surveys.

The recruitment procedures are restricted to adults that are of minimum age of 18 years old, must be currently seeing a therapist for the past six months, and they must be current users of ChatGPT. The recruitment method would be (1) upon logging in, users will see an ‘opt-in’ or ‘opt-out’ choice when using the therapist-like support GPT and they satisfy all the requirements. The incentive would be free ChatGPT Premium for the duration of the experiment which is disclosed when choosing if they want to opt-in or out.

The specific recruitment message I would display would be: **“We are conducting a voluntary research study to discover how ChatGPT can support people’s emotional well-being alongside professional therapy. If you are currently working with a professional therapist and would like to participate, you will receive ChatGPT Premium as compensation for your time during the duration of the experiment. Participation is completely optional and has no effect on any of ChatGPT’s services. ”**

The specific consent message I would display would be: **“By participating in this study you acknowledge that:**

- 1. ChatGPT is not a professional therapist and is a supplementary tool to be utilized alongside professional counsel.**
- 2. You may stop participating at any moment in time.**
- 3. Your responses may be recorded and analyzed for research purposes.**
- 4. Do not rely on ChatGPT for any professional or emergency support. In case of an emergency please contact a licensed professional.**
- 5. This tool is intended to better understand human feelings and the effectiveness of conversational AI as a supplement to therapy. This is not a form of medical or mental health treatment.**

I (insert signature here) consent to participating in this research study.”

My recruitment and consent procedures would be compatible with ethical requirements because it is explicitly informing the users of what exactly the experiment is doing, their use of it is monitored and analyzed, it is voluntary and can opt out of it at any time, and that it is a supplementary tool to be used alongside professional help and not a replacement. It’s completely transparent, upfront, and voluntary. Due to the nature of the pool of participants that already use therapists and ChatGPT as well, this is a form of selection bias, meaning this study only consists of participants with these requirements. There is also familiarity bias in which users are already familiar with the interface and might have a pre-conceived notion of ChatGPT already which might affect the way they respond in their surveys. One last bias this experiment might introduce is a positive incentive bias due to the nature of the incentive of free ChatGPT Premium which might affect their responses in the survey to being more positive.

5 REFERENCES

1. Belanger, A. (2026, January 15). ChatGPT wrote “Goodnight Moon” suicide lullaby for man who later killed himself. Ars Technica. <https://arstechnica.com/tech-policy/2026/01/chatgpt-wrote-goodnight-moon-suicide-lullaby-for-man-who-later-killed-himself/>